

Towards A Framework For Farm-scale Digital Twins

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ABSTRACT

Enhancing agricultural productivity while maintaining ecological balance amidst climate change is a looming challenge. The future of resilient farming and food security will depend upon the effectiveness of collecting, interpreting, and acting on data. An agricultural digital twin (DT) can provide a feedback loop which improves both farm management and the computer system which informs it through integrating right-time sensor data, process-based models (PBMs), data-driven models (DDMs), and hybrid approaches. Three demonstrator DTs for farm ecosystems are currently under development, utilizing extensive datasets from three instrumented research farms at the North Wyke Farm Platform in Devon, UK to drive and evaluate the accuracy of models in simulating key agroecosystem processes, such as soil nutrient cycling, water balance, and crop performance. The implementation process involves data collection, processing, model integration, and visualization. Key measurements are gathered up to every 15 minutes. PBMs along with DDMs and hybrid models will be utilized in an ensemble to enhance predictive accuracy and robustness. The DT architecture consists of three tiers. A client tier focuses on creating a user-friendly web frontend and API. An analysis and retrieval tier will facilitate the orchestration of services by a container registry and Kubernetes master node. A simulation tier will handle intensive data processing and model simulations with Apache Spark and high-performance computing nodes. We expect the DTs to improve decision-making, enhance system resilience against biotic and abiotic stresses, and pave the way for sustainable agricultural innovation.

CCS CONCEPTS

• Computing methodologies → Modeling and simulation.

KEYWORDS

Digital twin, Information Management Framework, modelling, agriculture

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1 INTRODUCTION

Mitigating food insecurity in the context of climate change is a major challenge of the 21st century [2]. To address this, it is necessary to maintain or enhance productivity from agriculture while balancing ecological priorities. This balance requires timely decisions on agronomic management, system conversion, and the adoption of specific farming philosophies. The goal is to increase yields, improve soil health, and reduce nutrient loss in response to the growing stresses of climate change and intensive farming practices..

For instrumented farms, modelling is an effective approach to support decision-making. Data-driven models (DDMs), such as Machine (and Deep) Learning models, can leverage large data sets collected from various sensors, including multi-spectral and hyper-spectral images, environmental data, historical data, and records of anomalies and events to predict outcomes and optimize resource use. Agroecosystem process-based models (PBMs), as mathematical models, have much thinner data requirements than DDMs and can simulate key processes such as soil nutrient cycling, water balance, and crop/livestock performance. They can be enriched with ground, proximal, and remotely sensed data. DDMs and PBMs can in turn inform the farm of recommended changes in management and data collection to improve both farm and model performance.

These two-way information flows between farm and models comprise a *digital twin* (DT) of the farm's agroecosystem. Set within an interactive visual environment, a farm DT enables insights and discoveries about the farm's past, current, and future status based on understanding the uncertainty around modeling forecasts. However, there has been very little development and application of agricultural DTs to date [10].

This short paper summarizes the methodology for creating three demonstrator farm DTs informed by the open and unique datasets of three instrumented research farms (grassland and arable) at Rothamsted Research's North Wyke campus in Devon, UK.



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2 METHODOLOGICAL FRAMEWORK

To develop and implement DTs effectively, a comprehensive methodological framework is required. The framework must integrate various data management techniques, modelling approaches, and analytical tools to ensure accurate simulations and associated measures of uncertainty. It should also be supported by the technical scaffolding necessary to ensure fast, efficient, and secure information transfer.

2.1 Information Management Framework

The complexity and scale of data involved in creating a DT necessitate a robust Information Management Framework (IMF) [11]. An IMF ensures that data are collected, processed, stored, and utilized efficiently, maintaining the integrity and security of the data throughout its lifecycle. An IMF also facilitates seamless integration and interaction between different data sources and models, enabling comprehensive analysis and decision-making. A well-designed IMF is crucial for managing the diverse and voluminous data streams that feed into a DT. It helps in standardizing data formats, ensuring data quality, and appropriate data storage. Furthermore, it supports real-time data processing and analytics, which are essential for the dynamic nature of DTs. An effective IMF can significantly enhance the performance and utility of DTs. For instance, an IMF facilitated the integration of heterogeneous data sources in a manufacturing DT, leading to improved predictive maintenance and operational efficiency [12]. Typically an IMF for a farm consists of seven levels:

- (1) Data Collection: Research farm measurements, recordings, events, external remote sensing products, external simulated climate scenarios.
- (2) Data Integration: Data ingestion, replication/backup, storage, and indexing; message queues and brokers.
- (3) DDM Analytics: Data preprocessing and transformation; diagnostic, predictive, and prescriptive analytics; machine/deep learning and statistical model toolbox.
- (4) PBM Analytics: Data requirements and integration, simulation engines, diagnostic and predictive analytics, development of emulators and hybrids with DDMs.
- (5) Forecasting and Updating: Scenario design; PBM, DDM, emulator, and hybrid forecasts; data assimilation tools; feedback to physical system.
- (6) Visualization: User interface, decision support system and visualization toolbox, interactive dashboards, mobile-friendly access.
- (7) Security and Privacy: Access control, encryption, authentication, anonymization, monitoring and intrusion detection, regulatory compliance, audit and logging.

2.2 Data Management

Four research farms in Devon, UK, are collectively known as the North Wyke Farm Platform (NWFP). Established in 2010 (<http://resources.rothamsted.ac.uk/farmplatform>), the platform has a rich history of PBM application and development [13], making it ideally suited for DT development and eventual transference of concepts and digital structures to real farms (Figure 1).

The three NWFP farms in this study are comprised of two grassland (grazing) and one arable (small grains cropping) systems,

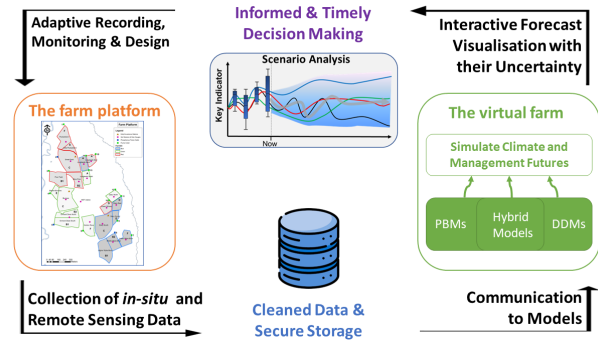


Figure 1: Process flow of a demonstrator Digital Twin with the North Wyke Farm Platform as a host data site.

equipped with a variety of sensors and data collection instruments that monitor such processes as soil condition, crop growth, live-stock performance, and nutrient cycling in soil, air, and water. The collected data are used to understand the intricate interactions within agroecosystems and to evaluate different farming practices. Key features of the NWFP include:

- **Data Collection:** Collect rich datasets for conducting longitudinal studies and observing the long-term impacts of different agricultural practices. Measurements from over 400 in situ sensors collected up to every 15 minutes provide input values to initialize, parameterize, and drive the models. Datasets are also used to validate and calibrate the models.
- **Data Processing:** The collected data is cleaned, transformed, and stored in a centralized Microsoft SQL Server database. This process involves removing noise and errors from the data to improve its utility.
- **Data Integration:** Data from multiple (internal and external) sources are integrated into a cohesive framework to support comprehensive analytics and DT decision-making.
- **Data Sharing:** Farm data are available for download through a portal and via a library of APIs for remote access by third-party software..

2.3 Modelling Approaches

Building on the NWFP infrastructure and data, the implementation of the farm-scale DTs involves the integration of several modelling approaches within the framework, including:

Process-Based Models (PBMs) are mathematical representations of natural systems that capture the underlying biological, physical, and chemical processes determining a system behaviour such as soil nutrient cycling, water balance, crop, and livestock performance. They are essential for creating accurate and reliable DTs due to their capacity to i) clarify complex relationships associated with natural systems, ii) investigate multiple farming scenarios and iii) quantify the different trade-offs related to agricultural practices. PBMs under current consideration are SPACSYS [14], developed at Rothamsted Research specifically with NWFP data, and includes an animal growth model; LandedDND [4], with sophisticated

soil and nutrient cycling processes; APSIM [7, 5], with highly refined crop growth processes; and RothC_Py [6, 1], also developed at Rothamsted Research, which specializes in soil organic carbon. Except for RothC_Py, all models include nutrient cycling and plant growth processes.

Data-Driven Models (DDMs) are derived using machine learning or statistical analysis to detect patterns, relationships, or trends in large datasets. DDMs excel in real-time decision making by continuously learning from incoming on-station and remotely sensed data making them highly adaptative to changing farm conditions. The DDMs under evaluation will be effective for predicting site specific outcomes (i.e., yield, soil moisture, soil-based emissions), setting thresholds for efficient land management and recommending optimal farming strategies.

Table 1: Modes of interaction between Process-Based Models (PBMs) and Data-Driven Models (DDMs) in the Digital Twin framework

Interaction	Mechanism	Description
Separate	Independent operation	PBMs and DDMs function independently.
Complementary	Feature generation	PBMs generate essential features that are not directly measurable, which DDMs then use as inputs to enhance prediction accuracy.
	Parameter estimation	DDMs estimate missing or uncertain PBM parameters.
Hybrid	Sequential integration	PBMs and DDMs work in sequence, where the output of one model feeds into the other.
	Parallel integration	PBMs and DDMs run independently in parallel, with their outputs combined through ensemble modelling to form a more accurate prediction.

Despite their benefits in agricultural decision-making, each approach has its limitations. PBMs, while being robust in representing mechanistic processes, can struggle with input uncertainties, and limitations in spatial and temporal data resolution. They often fail to capture the inherent variability of natural systems, which can restrict their ability to deliver precise predictions at local scales or over extended periods. Conversely, DDMs face limitations related to scalability, reliance on high-quality datasets, susceptibility to overfitting, and their inherent black-box nature. To mitigate some of these limitations and enhance the scalability of our DT framework, data assimilation techniques are introduced. These methods integrate field-based sensor data and remotely sensed data with PBMs and DDMs. This integration improves model accuracy, adaptability, and the ability to account for inherent uncertainties related to model parameters, inputs, initial conditions, and environmental variability. Table 1 illustrates the interaction between PBMs and

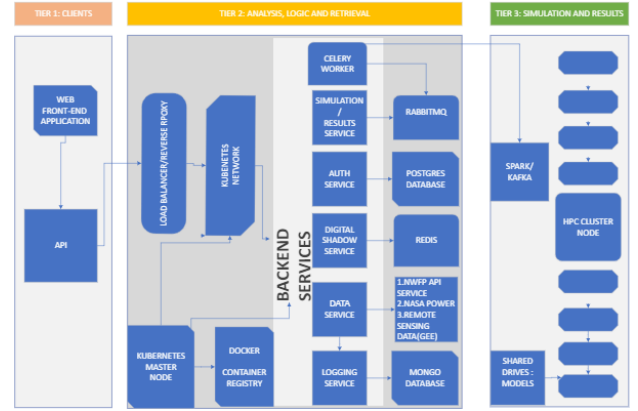


Figure 2: System Design and Architecture

DDMs within our DT framework, highlighting how these models can be used independently, complementarily, or in hybrid forms to assist agricultural decision-making.

2.4 From Field to Cloud: Digital Twin Architecture

Designing agricultural DTs requires addressing key factors like scalability, computational efficiency, adaptability, and interoperability to manage growing data from sources like soil sensors and climate models, while ensuring real-time processing and analysis.

Scalability is crucial to manage the growing data from agricultural activities like soil sensors and climate models, ensuring the system maintains performance for real-time processing and analysis. Computational efficiency is key, requiring optimized architecture and algorithms to quickly process large datasets with minimal resource use, enhancing DT responsiveness for real-time decisions. Adaptability is also vital, allowing the DT to integrate new data sources, adopt emerging technologies, and meet evolving user needs. Additionally, interoperability is essential for seamless integration with platforms like farm management software and IoT devices, providing a comprehensive view of farm operations to improve decision-making.

The architecture for the three NWFP-based Digital Twins (DTs) has been meticulously designed to emphasize the key factors mentioned earlier. This design ensures the system can manage increasing data volumes, accommodate concurrent user interactions, and deliver instantaneous, data-centric insights. Utilizing these features, the architecture facilitates informed decisions that aim to boost agricultural productivity and cut emissions. The system's framework is divided into three separate tiers, each focused on refining various facets of the DTs, thus ensuring the architecture's resilience, adaptability, and its ability to address the dynamic demands of agriculture.

- (1) Client Tier: This layer is responsible for user interaction with the DT system, offering interfaces to access, manipulate, and visualize DT data.
- (2) Analysis and Retrieval Tier: This middle layer retrieves and processes data from various sources, applying algorithms for

analysis and preparing the data for simulations or display to the user.

- (3) **Simulation and Results Tier:** This final layer handles simulations, predicting outcomes, optimizing operations, and generating actionable insights delivered to the client tier.

2.4.1 Tier 1: Clients.

- (1) **Web Frontend Application:** This component acts as the primary user interface for farmers, farm advisors and stakeholders. It enables users to input data, visualize outcomes, and access various tools and applications that facilitate decision-making. It enables users to easily interact with the DT system.
- (2) **The Application Programming Interface (API)** The API acts as a bridge between the web frontend and backend services, handling requests and retrieving data.

2.4.2 Tier 2: Analysis and Retrieval.

- (1) **Container Registry:** This component stores container images for deploying microservices, ensuring correct versions are available, supporting version control, and maintaining consistency.
- (2) **Load Balancer:** The load balancer distributes incoming network traffic across multiple instances of the backend services.
- (3) **Kubernetes Master Node:** Kubernetes manages the deployment, scaling, and operation of containerized applications, with the master node overseeing worker nodes to ensure proper scaling and performance.
- (4) **Backend Services:** These backend services Handles core business logic and data processing, with multiple instances of backend services to manage requests concurrently, enhancing system efficiency and reliability.
- (5) **RabbitMQ:** This message broker facilitates inter-service communication by queuing messages and ensuring they are delivered in the correct order. It supports asynchronous processing, allowing the system to manage tasks without delaying the main workflow.
- (6) **Redis:** An in-memory data structure store used primarily for caching and real-time data processing. Redis helps improve the performance and speed of the system by storing frequently accessed data in memory.
- (7) **Postgres Database:** A robust, scalable relational database, Postgres stores persistent data, ensures integrity, supports complex queries, and offers high availability.
- (8) **Celery Worker:** Celery workers execute asynchronous tasks and background jobs, such as data processing, sending notifications, or handling long-running computations. This ensures that these tasks do not block the main application flow.

2.4.3 Tier 3: Simulation and Results.

- (1) **Spark:** Apache Spark allows for distributed data processing, enabling the system to handle large volumes of data and perform complex simulations and analytics. Spark's capabilities include real-time stream processing and machine learning, which are crucial for generating actionable insights.

Table 2: Advantages and Trade-offs in the Architecture Components

Layer (Tier)	Component	Advantages	Trade-offs
Client Tier	Web Frontend	User-friendly interface. Intuitive interaction for end-users.	Requires continuous updates to ensure usability. Potential performance bottlenecks if not optimized.
	API Gateway (RESTful API)	Seamless communication between frontend and backend. Standardized interface.	May require additional security measures. Potential latency in high-traffic scenarios.
Analysis and Retrieval Tier	Load Balancer (Nginx or HAProxy)	Distributes traffic effectively. Ensures high availability.	Can be a single point of failure if not properly managed. Adds complexity to network configuration.
	Kubernetes	Efficient container orchestration. Scalability and flexibility.	Complexity in setup and management. Requires expertise in containerization.
	Backend Services (Microservices with Docker)	Loosely coupled services. Independent scaling and deployment.	Overhead of managing multiple services. Increased inter-service communication complexity.
Simulation and Results Tier	Data Processing (Apache Spark)	Handles large-scale data processing. Supports real-time analytics.	High resource consumption. Requires specialized knowledge for optimization.
	HPC Cluster Nodes	High computational power for simulations. Supports complex model calculations.	High cost in terms of infrastructure. Requires specialized knowledge for effective use.
	Simulation Modules (Custom-built)	Tailored to specific agricultural models. Highly customizable.	High development time and effort. Requires continuous maintenance and updates.

- (2) HPC Cluster Node: High-Performance Computing (HPC) nodes provide the computational power needed for intensive simulations and complex model calculations.
- (3) Simulation Modules: These modules run various simulation scenarios and models within the HPC via Spark, predicting outcomes, optimizing resources, and planning. Results are processed and made available to users on the client tier.

2.4.4 Microservices Architecture. Monolithic architectures, while simpler to implement initially, can become problematic as systems grow. They often present challenges in updating or modifying parts which affect the entire system, leading to potential downtime and limited flexibility. These issues can create bottlenecks in performance and scalability.

To address these issues, the selected architecture for the NWFP DTs is based on microservices[15], enabling each service to function independently. This approach offers key advantages: services are loosely coupled, allowing for independent development, deployment, scaling, and updates. This modularity enhances system flexibility, scalability, and agility in responding to changes. Additionally, microservices support continuous integration and deployment, reducing downtime and improving system resilience.

Adopting a microservices architecture is particularly advantageous for complex systems like the NWFP DTs, where independently managed components boost efficiency and reliability. This modular design allows for flexible integration of legacy changes, maintaining backward compatibility so that older components can coexist seamlessly with new ones. The architecture supports incremental updates, enabling the system to evolve gradually without a complete overhaul, preserving valuable legacy elements. This approach ensures operational continuity, enhances resilience, and strengthens the NWFP's adaptability, keeping it effective and relevant in the face of ongoing changes.

2.5 Interactive Farm Digital Twins

An interactive user interface is crucial for leveraging Digital Twins (DTs) in agricultural decision-making. It should integrate high-resolution maps and real-time data streams to support decisions on water usage, fertilizer application, and stocking density, optimizing resource management. [3, 9]. The interface must allow for real-time exploration of field sensors and remotely sensed data, supporting efficient farm operations with immediate insights [3].

Users can apply various simulation models—data-driven, process-based, and hybrid—tailored to each farm's needs, to run targeted scenario simulations and explore sustainability strategies [8, 10]. Predictions must be presented with quantified uncertainty, aiding in accurate forecasting and data-driven planning, ultimately helping optimize yields while minimizing environmental impacts [8, 9].

Designing an effective farm Digital Twin interface requires precise capture of physical farm details, ensuring an accurate and user-friendly digital representation. This tool aids farmers and stakeholders in decision-making, efficiency, and farm management. Key considerations, as shown in Figure 3, include:

- **Detailed Mapping and Visualization:**
 - **High-Resolution Maps:** Use high-resolution satellite or drone imagery to create detailed maps of the farm (Figure 3.a).

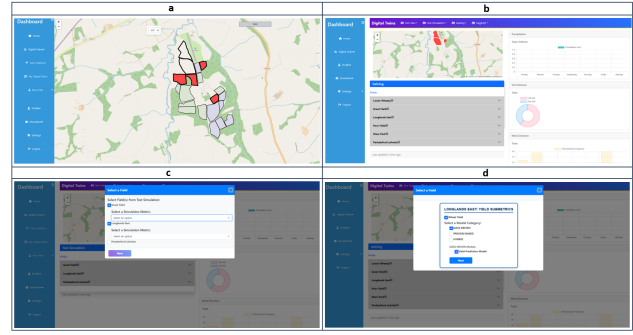


Figure 3: Interactive User Interface for Farm Digital Twin

- **Interactive Elements:** Allow users to click on different parts of the map to get detailed information about specific fields or assets (Figure 3.d).
- **Real-Time Data Integration:**
 - **Sensor Data:** Integrate data from various sensors placed around the farm, including soil moisture sensors, weather stations, and crop health monitors.
 - **Live Updates:** Ensure the interface handles real-time data updates, delivering the most current information, as demonstrated by the precipitation and soil moisture widgets in Figure 3.b.
- **Simulation and Predictive Analytics:**
 - **Yield Predictions:** Offer predictive analytics for yield based on current and historical data, similar to the yield submetrics selection in Figure 3.d.
- **User-Friendly Interface:**
 - **Dashboard Design:** Create a dashboard that is intuitive and easy to navigate, with customizable widgets that display key metrics, as seen in Figure 3.c.
- **User Interaction and Feedback:**
 - **Collaboration Tools:** Include tools for collaboration, enabling multiple users to work together and share insights.
 - **Feedback Integration:** Implement a continuous feedback loop that integrates real-time data and user input into the DT. This loop refines the DT's models, ensuring its predictions align closely with actual farm conditions. By regularly updating based on user experiences and system performance, the DT adapts and evolves to support effective decision-making.

The current development of our DT includes both technical advancements and stakeholder engagement. Currently, SPACSYS and RothC_Py are integrated in the DT to simulate soil carbon dynamics. APIs to retrieve data from soil moisture, meteorological, and eddy covariance stations in NWFP are implemented. A sensitivity analysis of LandscapedDNDC is being undertaken and Google Earth Engine (GEE) APIs are implemented to process satellite imagery, extract vegetation maps, and fill gaps in primary soil moisture and weather datasets. Research dissemination is being carried out through onsite workshops, where key components of the DT are being discussed and demonstrated, ensuring effective implementation across various agricultural settings.

3 DISCUSSION AND NEXT STEPS

As described, demonstrator DTs are in development for three research farms. The component processes (soils, crops, livestock, biodiversity, and water/gaseous emissions) of the farms are multi-scale with complex space-time and above-below ground connections. Respective component measurements (sensed or sampled) in space and time will be used to update PBM simulations at the appropriate scale of decision-making. Given improved capture of error and uncertainty, the risk of poor farm management decisions can be quantified, facilitating planning for sustainable farming in the long term, coupled with short-term early warning signals for diminishing system resilience to increasing threats of abiotic and biotic stresses.

Success will also depend upon the ability of the DT to effectively address a number of challenges. For example, data collection requires instrument calibration and accurate transfer into the holding database. Manual data review and cleaning are required to ensure appropriate quality, delaying the release of data. This demonstrator DT has the particular challenge of being developed on a research farm where management in each field is relatively fixed and inflexible due to the specific research questions for which they were originally developed, predating the DT. To address this, this demonstrator DT will be calibrated and run for the earliest years of the experimental period with "future" predictions made over the most recent years. This has the advantage of immediate verification of predictions as the "future" has already taken place. As DT framework matures, future changes in management onsite may include farm fields dedicated to dynamic DT feedback, but regardless, verification of the effectiveness of management changes requires timescales of days to months (in the case of greenhouse gas emissions and water quality) to years (in the case of soil organic carbon).

All farm agroecosystems are inherently complex in space, time, and scale, where sensing technologies for most processes are either limited or non-existent. Thus, achieving a fully data-driven model characterising the entire farm ecosystem is currently unfeasible. The PBM ensemble also presents the challenge of increasing data requirements as more models are added. Each PBM has its own approach to simulating the farming system and while there are many common input elements required, each model also has its own unique requirements and units of values. This heterogeneity complicates the integration and synchronisation of multiple models. PBMs also require relatively long run times and involve calibration which introduces a layer of complex automation necessitating high computing power.

The architecture supporting the DTs faces significant challenges, particularly in integrating legacy data sources with the new DT framework, as many existing systems were not designed for the dynamic data flows that DTs require. Ensuring seamless data interoperability and real-time processing is essential for the effective functioning of the DT. Data availability and accessibility are key concerns. Another major issue is managing missing data, common in real-world collection due to sensor malfunctions or transmission errors. Missing values can compromise DT simulations' accuracy, especially in complex systems where precision is crucial. Statistical

or machine learning techniques can impute these gaps, but this introduces uncertainty, as imputed values are estimates.

Once deployed and tested, our aim is to extend the DT architecture of the demonstrator DTs beyond experimental farms to real farms, provisioning each with its own bespoke DT depending on its level of sensing capacity. Contrasting the operations and productivity of the farms against the DT recommendations is a target research activity.

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